

DULUTH SKY HARBOR, MN, USAWMO: **727456**Lat: **46.722N**Lon: **92.043W**Elev: **186**StdP: **99.11**Time Zone: **-6.00 (NAC)**Period: **98-19**WBAN: **04919****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.8	(c) -21.4	(d) -28.8	(e) 0.3	(f) -22.8	(g) -27.2	(h) 0.3	(i) -21.1	(j) 12.7	(k) -4.0	(l) 11.5	(m) -3.9	(n) 3.6	(o) 220	(p) 0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.8	(c) 30.0	(d) 21.9	(e) 27.9	(f) 20.7	(g) 27.0	(h) 20.0	(i) 23.9	(j) 28.0	(k) 22.5	(l) 26.3	(m) 21.2	(n) 24.8	(o) 5.4	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.6	26.1	21.3	16.3	25.0	19.8	14.8	23.6	72.6	28.0	67.1	25.8	62.3	24.0	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.8	(b) 11.3	(c) 9.7	(e) -27.1	(f) 33.3	(g) 2.5	(h) 2.1	(i) -28.9	(j) 34.8	(k) -30.3	(l) 36.0	(m) -31.7	(n) 37.2	(o) -33.5	(p) 38.7		
			DB													
			WB													
				-27.1	25.9	2.2	1.7	-28.7	27.1	-30.0	28.1	-31.3	29.1	-32.9	30.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-8.9	-7.5	-1.6	4.0	9.2	14.6	20.2	19.4	15.2	8.4	1.6	-5.9
	DBStd	11.08	6.94	6.19	5.56	3.93	4.16	4.37	3.68	3.18	4.02	4.03	5.18	6.35
	HDD10.0	2527	587	490	360	184	67	8	0	0	4	78	254	494
	HDD18.3	4751	846	723	618	429	284	128	24	25	111	308	502	753
	CDD10.0	991	0	0	0	5	43	146	316	291	160	29	2	0
	CDD18.3	172	0	0	0	0	1	15	82	57	16	1	0	0
	CDH23.3	1293	0	0	0	1	36	150	653	343	104	6	0	0
	CDH26.7	362	0	0	0	0	7	36	207	87	23	1	0	0
Wind	WSAvg	4.1	4.0	4.1	3.9	4.7	4.2	3.8	3.6	3.6	4.1	4.5	4.4	4.2
Precipitation	PrecAvg	774	23	23	36	65	78	107	112	93	93	82	46	29
	PrecMax	990	53	54	78	222	143	170	254	192	185	177	114	59
	PrecMin	535	2	4	11	16	28	35	46	33	25	19	1	4
	PrecStd	114	15	13	19	46	29	39	51	42	43	41	33	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	7.6	12.9	21.3	27.5	30.9	32.8	32.2	29.0	22.9	16.2	7.2
		MCWB	2.8	4.4	7.0	11.4	17.0	23.5	24.2	24.9	22.0	15.7	10.4	4.2
	2%	DB	3.0	3.9	9.0	16.4	22.8	27.4	31.0	28.8	26.4	18.9	12.4	4.2
		MCWB	1.7	2.0	5.2	9.2	14.3	18.8	22.4	22.0	20.2	14.0	8.7	2.9
	5%	DB	2.1	2.4	7.0	12.6	19.0	24.7	28.8	27.2	23.6	17.0	10.2	3.0
		MCWB	1.0	1.1	3.9	7.3	13.1	17.6	21.4	20.5	18.2	12.6	7.5	2.1
	10%	DB	0.9	1.2	4.1	9.0	16.3	22.3	27.4	25.0	21.4	13.8	8.0	2.3
		MCWB	-0.1	0.1	2.2	5.6	11.3	17.0	20.5	19.5	17.4	10.8	6.4	1.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	5.0	8.6	12.8	18.5	23.5	25.8	25.8	23.7	17.9	11.7	5.0
		MCDB	5.3	7.5	11.6	17.8	24.5	28.6	30.1	29.4	27.6	20.9	14.6	6.5
	2%	WB	2.0	2.6	5.7	10.2	15.7	20.5	24.0	23.8	21.3	15.1	9.6	3.5
		MCDB	3.2	3.7	8.6	15.8	21.4	24.2	28.5	27.4	25.0	18.0	11.9	4.4
	5%	WB	0.7	1.3	4.2	8.0	13.9	18.8	22.7	22.3	19.5	12.9	8.0	2.6
		MCDB	1.8	2.2	6.4	11.8	18.5	23.5	26.9	25.3	22.6	15.4	9.2	3.3
	10%	WB	-0.6	-0.1	2.9	6.1	11.9	17.4	21.5	20.7	18.0	11.7	7.0	1.5
		MCDB	0.3	1.0	4.2	8.8	15.2	21.1	25.7	23.6	19.9	13.5	8.4	2.3
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.6	7.5	6.5	7.5	8.1	8.8	7.9	7.8	6.6	6.2	6.6
		MCDBR	7.7	8.8	9.9	13.5	15.2	14.2	12.7	11.5	11.7	10.9	9.2	6.3
		MCWBR	6.5	7.1	6.7	7.8	8.4	7.6	6.1	5.9	6.5	6.7	6.4	5.4
	5% WB	MCDBR	7.1	7.7	8.9	12.5	13.7	12.6	11.3	9.8	10.2	9.1	8.1	5.8
		MCWBR	6.2	6.4	6.3	7.8	8.5	7.9	6.5	6.2	6.4	6.5	6.2	5.1
Clear-Sky Solar Irradiance	taub		0.250	0.274	0.317	0.356	0.386	0.383	0.388	0.393	0.371	0.318	0.292	0.270
	taud		2.412	2.355	2.353	2.362	2.331	2.370	2.378	2.385	2.417	2.551	2.512	2.417
	Ebn at Noon		866	906	909	899	879	882	871	852	841	848	805	791
	Edh at Noon		67	88	104	114	122	118	116	111	98	74	63	61
All-Sky Solar Radiation	RadAvg		1.44	2.52	3.68	4.59	5.33	5.77	6.12	5.12	3.85	2.34	1.40	1.02
	RadStd		0.15	0.23	0.34	0.46	0.39	0.40	0.25	0.36	0.26	0.30	0.16	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page